

Assignment 03: Data Classification using the Keras Functional API

CSci 560: Neural Networks and Deep Learning

Description

Welcome to our third assignment for the class. Its time to build your first neural network using the Keras API. You will be using a dataset to perform classification that only has 2 features, but that is not linearly separable. You will start with a simple single layer, which is equivalent to a simple logistic regressor, then you will increase the power of the model to try and see how that effects the decision boundary that a model with more representational power can make.

Instructions:

- Do not use loops (for/while) in your code, unless the instructions explicitly ask you to do so.
- Use the Keras functional API to implement your asked for network architectures, and to practice and get used to specifying models using that API.

Objectives

You will learn:

- How to implement a 2-class classification neural network with a single hidden layer
- How representational power effects the types of decision boundary a model can form for a classification problem.

Overview and Setup

Assignment Tasks

Assignment Submission

Additional Information